

Problem 1 (Linear Regression) Let's implement a tool needed for a computer vision algorithm called photometric stereo, which allows us to reconstruct a 3D object's surface from 2D images of it under different lighting conditions. Figure 1 gives a sneak peek at what you'll be able to do after completing all the pieces. We want to measure the 3D surface reconstruction shown in Figure 1 right. We model the object's surface as $z = f(x, y) \in \mathbb{R}_+$ where $x, y \in \mathbb{R}$ denotes coordinates on the table, and z is the height of the surface above the table. To measure the object, we change the light source orientation $\mathbf{l} \in \mathbb{R}^{3 \times 1}$ (a unit-norm vector). For each orientation $\mathbf{l}$, we could obtain a pixel intensity image $I(x, y)$. Under a diffuse reflection assumption, the following relationship holds:

$$I(x, y) = \alpha(x, y) \mathbf{l}^T \mathbf{n}(x, y), \quad \text{(1)}$$

where $\alpha(x, y) \in \mathbb{R}$ is related to the surface's own property and $\mathbf{n}(x, y) \in \mathbb{R}^3$ is the unit-norm surface normal vector of $f(x, y)$. Assume that for each coordinate (x, y) in $f(x, y)$, it is corresponding to a pixel (x, y) in an $I(x, y)$. Also assume the light is directional light, i.e., the orientation $\mathbf{l}$ for the light is the same for all pixels (x, y) .

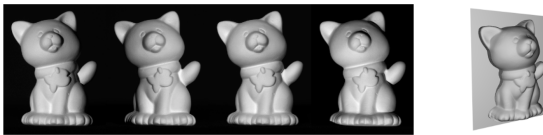


Figure 1: Photometric stereo. Left: 2D-pixel intensity $I(x, y)$ of a common scene under different lighting conditions. Right: 3D surface reconstruction $f(x, y)$ computed from the input images.

- (a) Suppose moving the light source in $\mathbf{l}_1, \dots, \mathbf{l}_d$, d different directions and obtain $\mathbf{I}_1, \dots, \mathbf{I}_d \in \mathbb{R}_+^{M \times N}$ 2D-pixel intensities matrix. We want to find the surface normal direction matrix $\mathbf{N} \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

We know $A \in \mathbb{R}^{m \times n}$, observation $y \in \mathbb{R}^m$
 We need to find a vector $x \in \mathbb{R}^n$ such that:

$$\begin{array}{c} Ax = y \\ \circ \quad \circ \quad \circ \end{array} \Rightarrow \underset{x}{\operatorname{argmin}} \frac{\|Ax - y\|_2^2}{\substack{\Downarrow \\ > 0}}$$

$$\Rightarrow x^* = A^+ y$$

A^+

(a). $p(x, y) \triangleq \underline{d(x, y)} \cdot \underline{n(x, y)} \in \mathbb{R}^3$

$\Rightarrow I_i(x, y) = l_i^T p(x, y)$ for $i = 1, \dots, d$ $y = Ax$

$d \left\{ \begin{array}{l} I_1(x, y) = l_1^T p(x, y) \\ I_2(x, y) = l_2^T p(x, y) \\ \vdots \\ I_d(x, y) = l_d^T p(x, y) \end{array} \right. \xRightarrow{\text{stack}} \begin{bmatrix} I_1(x, y) \\ I_2(x, y) \\ \vdots \\ I_d(x, y) \end{bmatrix} = \begin{bmatrix} l_1^T \\ l_2^T \\ \vdots \\ l_d^T \end{bmatrix} \cdot p(x, y)$

$y = Ax \xRightarrow{\text{stack}}$

$\Rightarrow \underline{b = L^T \cdot p}$ for any (x, y)

From pixel to the whole image matrix:

stack for each x, y we have $\begin{array}{ccc} \downarrow & & \\ b(x, y) = & L^T \cdot & p(x, y) \\ \in \mathbb{R}^d & \in \mathbb{R}^{d \times 3} & \in \mathbb{R}^3 \end{array}$

$\Rightarrow \underline{b^T(x, y)} = p^T(x, y) \cdot L$

$\downarrow$

$$\begin{bmatrix} b^T(1,1) \\ b^T(1,2) \\ b^T(1,N) \\ b^T(2,1) \\ \vdots \\ b^T(m,N) \end{bmatrix} = \begin{bmatrix} \rho^T(1,1) \\ \rho^T(1,2) \\ \vdots \end{bmatrix} \cdot L \quad L \in \mathbb{R}^{3 \times d}$$

$$\underline{B}^T \in \mathbb{R}^{mN \times d} \quad \underline{\Omega}^T \in \mathbb{R}^{mN \times 3}$$

$$\begin{cases} b^T(1,1) = \rho^T(1,1) \cdot L \\ b^T(1,2) = \rho^T(1,2) \cdot L \\ b^T(m,N) = \rho^T(m,N) \cdot L \end{cases}$$

where $B = [b(1,1), \dots, b(m,N)] \in \mathbb{R}^{d \times mN}$

$\Omega = [\rho(1,1), \dots, \rho(m,N)] \in \mathbb{R}^{3 \times mN}$

$L = [l_1, \dots, l_d] \in \mathbb{R}^{3 \times d}$

$$\Rightarrow B^T = \Omega^T \cdot L$$

$$\Rightarrow \underline{B} = \underline{L}^T \cdot \underline{\Omega}$$

$$\Rightarrow \underline{\Omega}^* = \underset{\Omega}{\operatorname{argmin}} \|\underline{B} - \underline{L}^T \underline{\Omega}\|_F^2$$

Finally: $n(x,y) = \frac{\underline{\Omega}^*(x,y)}{\|\underline{\Omega}^*(x,y)\|_2}$, since $\|n(x,y)\|_2 = 1$

(b) Recovering the surface matrix $z \in \mathbb{R}_+^{M \times N}$ from the surface normal direction matrix $N \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

Hint 1: The relationship between surface norm $n(x,y)$ and the object surface function $f(x,y)$ is:

$$\underline{n}(x,y) = \frac{\left[-\frac{\partial \underline{z}(x,y)}{\partial x}, -\frac{\partial \underline{z}(x,y)}{\partial y}, 1 \right]^T}{\sqrt{1 + \left(\frac{\partial \underline{z}(x,y)}{\partial x} \right)^2 + \left(\frac{\partial \underline{z}(x,y)}{\partial y} \right)^2}}$$

$$\underline{z}(x,y)$$

Hint 2: The $\underline{z}(x,y)$ could also be viewed as an image, such that:

$$\frac{\partial \underline{z}(x,y)}{\partial x} \approx \frac{\underline{z}(x+1,y) - \underline{z}(x,y)}{(x+1) - x} = \underline{z}(x+1,y) - \underline{z}(x,y)$$

$$\frac{\partial \underline{z}(x,y)}{\partial y} \approx \frac{\underline{z}(x,y+1) - \underline{z}(x,y)}{(y+1) - y} = \underline{z}(x,y+1) - \underline{z}(x,y)$$

(b). We know $N \in \mathbb{R}^{3 \times m \times n}$, we need to find $Z \in \mathbb{R}^{m \times n}$

First find the relation between N and Z

$$\frac{n_1(x, y)}{n_3(x, y)} = \frac{\partial Z(x, y)}{\partial x} = Z(x+1, y) - Z(x, y)$$

$$\frac{n_2(x, y)}{n_3(x, y)} = \frac{\partial Z(x, y)}{\partial y} = Z(x, y+1) - Z(x, y)$$

$$\Rightarrow \begin{cases} N_{13}(x, y) = Z(x+1, y) - Z(x, y) \\ N_{23}(x, y) = Z(x, y+1) - Z(x, y) \end{cases} \text{ for each } x, y$$

$$\Rightarrow \begin{cases} N_{13} = D_1 Z \\ N_{23} = D_2 Z \end{cases} \Rightarrow \begin{bmatrix} N_{13} \\ N_{23} \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \end{bmatrix} Z$$

$$N = \begin{bmatrix} D_1 \\ D_2 \\ 0 \end{bmatrix} Z$$